

Applications of Differential Calculus (Related Rates Part II)

Name
Date
Course

Differentiation Formulas

Constant Rule i) $\frac{d}{dx}(c) = 0$ "c" is a constant.

Power Rule ii) $\frac{d}{dx}(x^n) = nx^{n-1}$ "n" is a constant.

Constant Multiple Rule iii) $\frac{d}{dx}[cf(x)] = cf'(x)$ "c" is a constant.

Sum Rule iv) $\frac{d}{dx}[f(x) + g(x)] = f'(x) + g'(x)$

Difference Rule v) $\frac{d}{dx}[f(x) - g(x)] = f'(x) - g'(x)$

Product Rule vi) $\frac{d}{dx}[f(x)g(x)] = f'(x)g(x) + f(x)g'(x)$

Quotient Rule vii) $\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}$

The Chain Rule viii) If $h(x) = f(g(x))$, then

$$h'(x) = f'(g(x))g'(x)$$

Rule for Exponential Functions ix) $\frac{d}{dx}(a^x) = a^x \ln a$, "a" is a constant and $a > 0$.

Rule for base "e" Exponential Functions x) $\frac{d}{dx}(e^x) = e^x$

Rule for Logarithmic Functions xi) $\frac{d}{dx}(\log_a x) = \frac{1}{x \ln a}$, "a" is a constant and $a > 0$.

Rule for the Natural Logarithm Function xii) $\frac{d}{dx}(\ln x) = \frac{1}{x}$

Rule for Composite Natural Logarithm Functions xiii) $\frac{d}{dx} \{ \ln[g(x)] \} = \frac{g'(x)}{g(x)}$

Trigonometric Differentiation Formulas

I) $\frac{d}{dx}(\sin x) = \cos x$ II) $\frac{d}{dx}(\cos x) = -\sin x$

III) $\frac{d}{dx}(\csc x) = -\csc x \cot x$ IV) $\frac{d}{dx}(\sec x) = \sec x \tan x$

V) $\frac{d}{dx}(\tan x) = \sec^2 x$ VI) $\frac{d}{dx}(\cot x) = -\csc^2 x$

Procedure for Solving Related Rates Problems

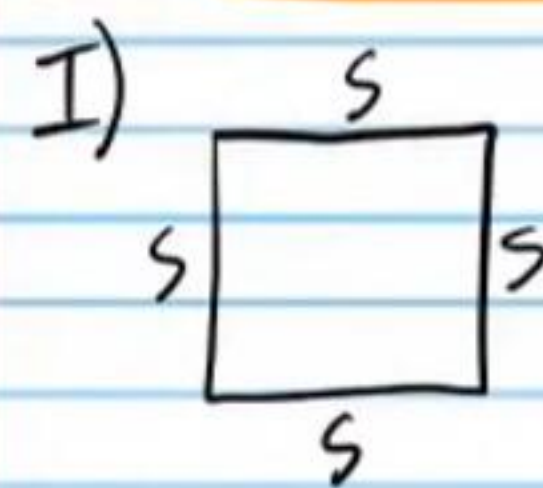
I) Draw a picture and label variables and constants. Some problems may require only a simple drawing.

II) Find an equation relating the variables in question. In many cases, some work must be done to write the equation using only two variables. In some rare cases, the problem can be solved when there is more than two variables in the equation.

III) Differentiate both sides of the equation, remembering to use the chain rule given that each variable is dependent (usually on time, sometimes on other things).

IV) Plug in known values into the equation (some additional math may be required to find all necessary values), and then solve for the unknown value.

- ① The lengths of all edges of a square are increasing by 5 ft/min. How fast is the area of the square increasing when its side lengths are each 3 ft?



II) $A = LW$
 $A = s \cdot s$
 $A = s^2$

III) $A = s^2$
 $\frac{dA}{dt} = 2s \frac{ds}{dt}$

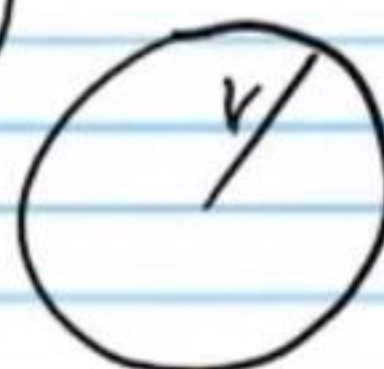
$$\frac{dA}{dt} = 2s \frac{ds}{dt}$$

IV) $\frac{dA}{dt} = 2s \frac{ds}{dt}$

$$\frac{dA}{dt} = 2(3)(5)$$

$$\frac{dA}{dt} = 30 \text{ ft}^2/\text{min}$$

- ② A snowball melts so that its volume decreases by $2 \text{ cm}^3/\text{h}$. What is the rate that its radius is decreasing when its radius is 1 cm ?

I)  II) $V = \frac{4}{3}\pi r^3$ III) $V = \frac{4}{3}\pi r^3$

$$\frac{dV}{dt} = \frac{4}{3}\pi (3r^2 \frac{dr}{dt})$$

$$\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$$

IV) $\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$

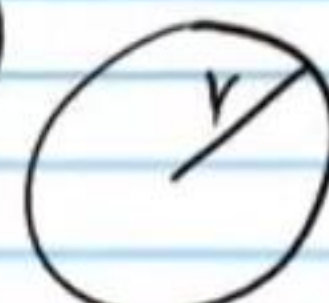
$$-2 = 4\pi (1)^2 \frac{dr}{dt}$$

$$-2 = 4\pi \frac{dr}{dt}$$

$$\boxed{\frac{dr}{dt} = -\frac{1}{2\pi} \text{ cm/h}}$$

$$A = \pi r^2$$

- ③ A drop of oil is added to a cup of water and the oil forms a growing circle on the surface of the water. If the area of the circle increases by $4 \text{ mm}^2/\text{s}$ when its radius is 11 mm , how fast does the radius of the circle increase at this moment?

I)  II) $A = \pi r^2$ III) $A = \pi r^2$

$$\frac{dA}{dt} = \pi (2r \frac{dr}{dt})$$

$$\frac{dA}{dt} = 2\pi r \frac{dr}{dt}$$

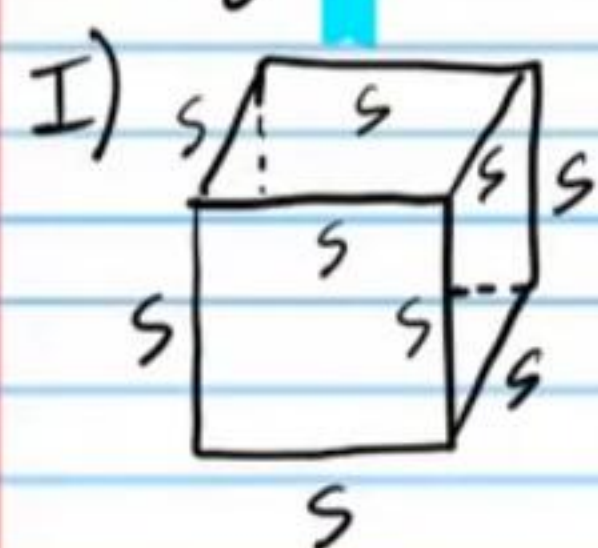
IV) $\frac{dA}{dt} = 2\pi r \frac{dr}{dt}$

$$4 = 2\pi (11) \frac{dr}{dt}$$

$$\boxed{\frac{dr}{dt} = \frac{2}{11\pi} \text{ mm/s}}$$

$$V = LWH$$

④ If the side length of a cube is decreasing by $\frac{1}{2}$ in/min when its side length is 5 in, what is the rate of change of the volume of the cube at this moment?



II) $V = LWH$ III) $V = s^3$

$$V = s \cdot s \cdot s$$

$$V = s^3$$

$$\frac{dV}{dt} = 3s^2 \frac{ds}{dt}$$

IV) $\frac{dV}{dt} = 3s^2 \frac{ds}{dt}$

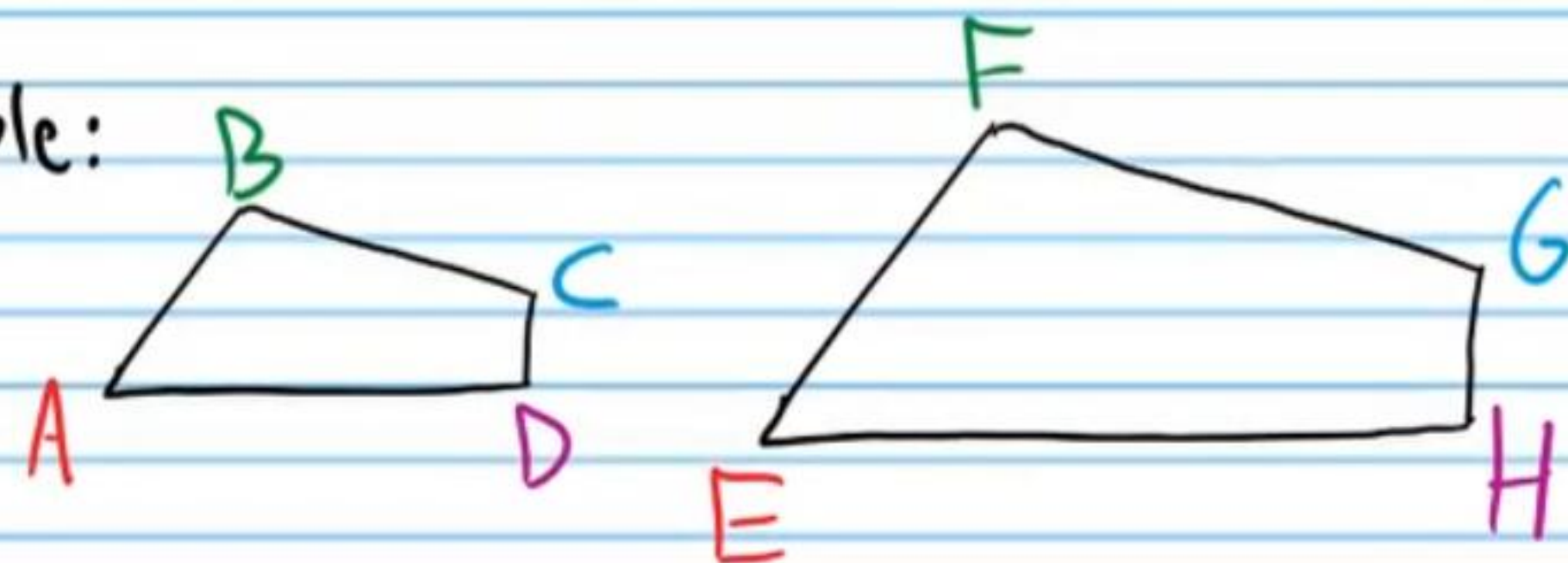
$$\frac{dV}{dt} = 3(5)^2 \left(-\frac{1}{2}\right)$$

$$\frac{dV}{dt} = -\frac{75}{2} \text{ in}^3/\text{min}$$

Review on Similar Polygons

Similar Polygons: Polygons with the same shape, but not always the same size. More specifically, polygons with congruent corresponding angles and equal ratios of corresponding sides (i.e. corresponding sides are proportional).

Example:

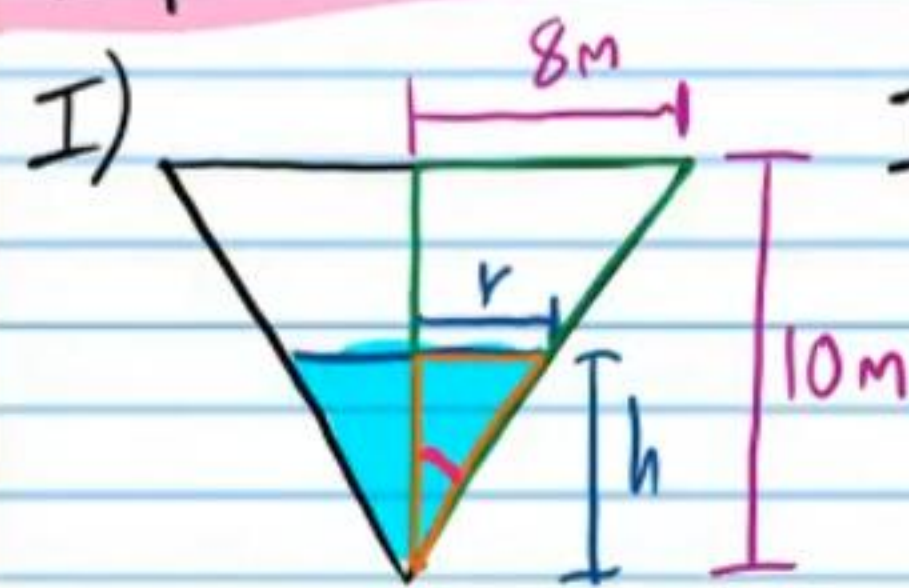


$$\angle A \cong \angle E, \angle B \cong \angle F, \angle C \cong \angle G, \angle D \cong \angle H$$

$$\frac{AB}{EF} = \frac{BC}{FG} = \frac{CD}{GH} = \frac{DA}{HE}$$

Angle-Angle Similarity Postulate: If two angles in one triangle are congruent to two angles in another triangle, then the triangles are similar. The postulate is abbreviated AA.

- ⑤ A water tank has the shape of a circular cone and the tip of the cone is the bottom of the tank. The radius of the top of the tank is 8 m and the height of the tank is 10 m. If 6 m^3 of water are poured into the tank per minute, find the rate that the water level is rising when the water is 3 m deep.



II) $V = \frac{1}{3}\pi r^2 h$

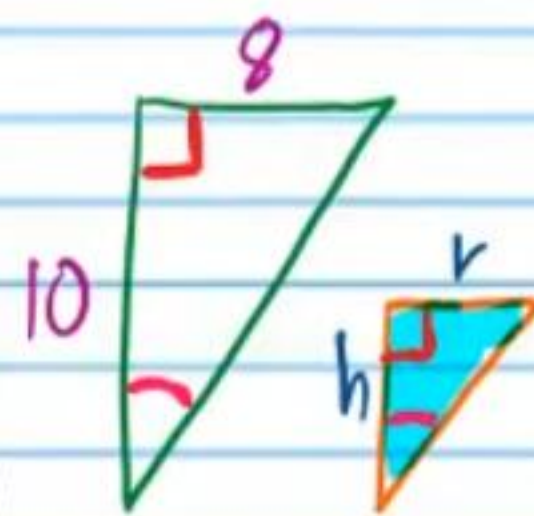
$$V = \frac{1}{3}\pi \left(\frac{4}{5}h\right)^2 h$$

$$V = \frac{1}{3}\pi \frac{16}{25} h^2 \cdot h$$

$$V = \frac{16\pi}{75} h^3$$

$$\frac{r}{8} = \frac{h}{10}$$

$$r = \frac{4}{5}h$$



III) $V = \frac{16\pi}{75} h^3$

$$\frac{dV}{dt} = \frac{16\pi}{75} \left(3h^2 \frac{dh}{dt}\right)$$

$$\frac{dV}{dt} = \frac{16\pi}{25} h^2 \frac{dh}{dt}$$

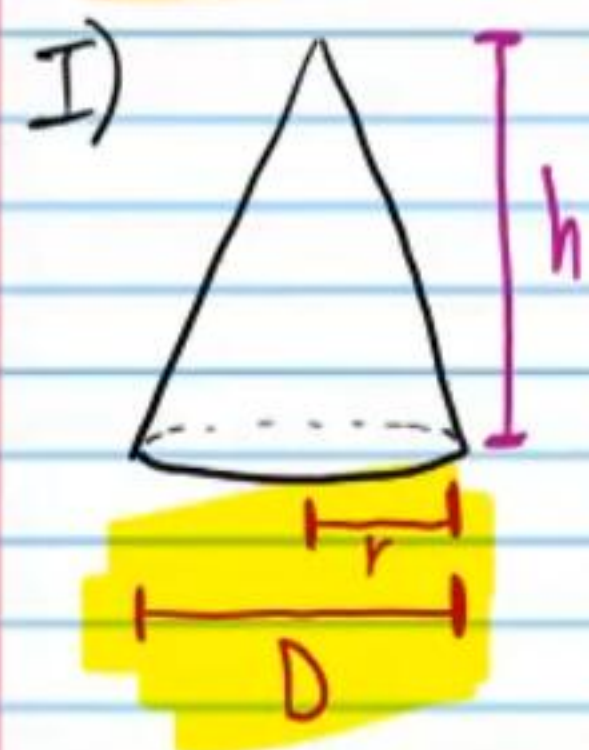
IV) $\frac{dV}{dt} = \frac{16\pi}{25} h^2 \frac{dh}{dt}$

$$6 = \frac{16\pi}{25} (3^2) \frac{dh}{dt}$$

$$6 = \frac{16\pi}{25} (9) \frac{dh}{dt}$$

$$\frac{dh}{dt} = \frac{25}{24\pi} \text{ m/min}$$

- ⑥ Sand is poured onto the ground creating a pile in the shape of a circular cone. The height of the pile is always triple the diameter of its base. If the diameter of its base is increasing by 2.7 feet/min, what volume of sand is being poured per minute when the radius of the base of the pile is 4 feet?



II) $V = \frac{1}{3}\pi r^2 h$

$$V = \frac{1}{3}\pi \left(\frac{D}{2}\right)^2 (3D)$$

$$V = \frac{1}{3}\pi \frac{D^2}{4} (3D)$$

$$V = \frac{\pi}{4} D^3$$

$$h = 3D$$

$$D = 2r$$

$$\frac{D}{2} = r$$

III) $V = \frac{\pi}{4} D^3$

$$\frac{dV}{dt} = \frac{\pi}{4} (3D^2 \frac{dD}{dt})$$

$$\frac{dV}{dt} = \frac{3\pi}{4} D^2 \frac{dD}{dt}$$

IV) $\frac{dV}{dt} = \frac{3\pi}{4} D^2 \frac{dD}{dt}$

$$\frac{dV}{dt} = \frac{3\pi}{4} 8^2 \cdot 2.7$$

$$\frac{dV}{dt} = 48\pi \frac{27}{10}$$

$$\frac{dV}{dt} = \frac{648\pi}{5} \text{ ft}^3/\text{min}$$

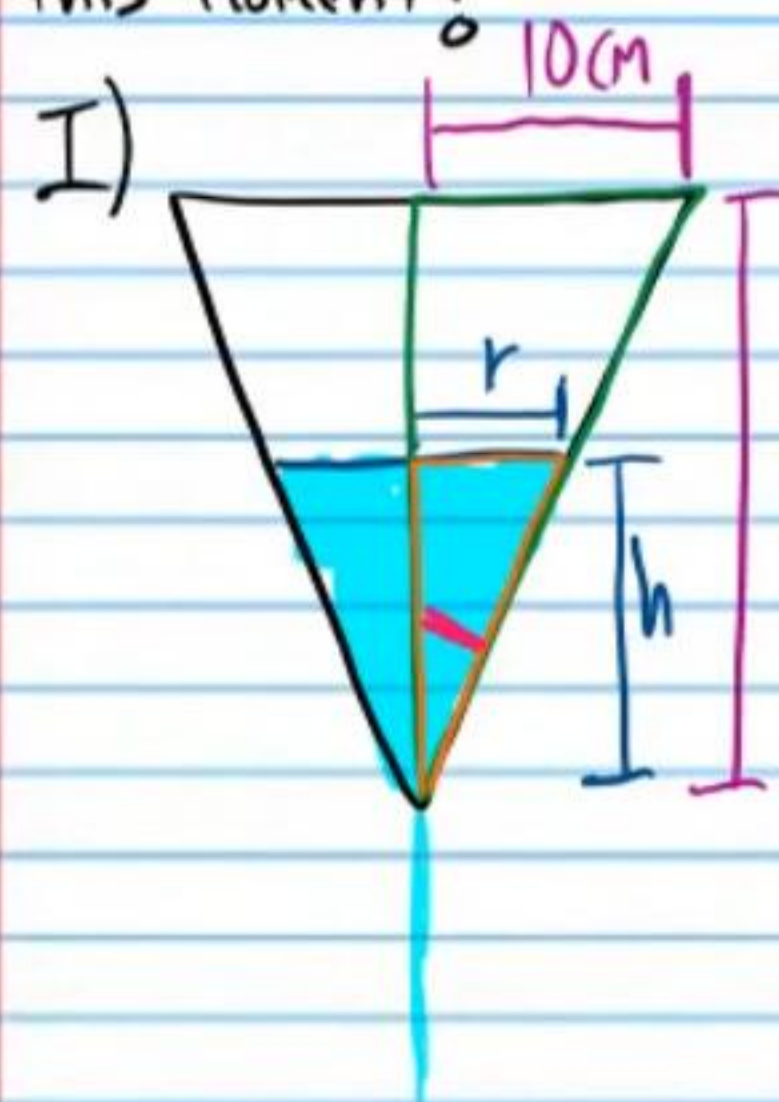
$$D = 2r$$

$$D = 2(4)$$

$$D = 8$$

* Your rates will be negative.

⑦ Water is leaking from an inverted conical tank. The height of the tank is 60 cm and the radius of the top of the tank is 10 cm. If the water level is decreasing at a rate of 0.7 cm/s when the water level is 30 cm, how much water is leaking out of the tank per second at this moment?



I) $V = \frac{1}{3}\pi r^2 h$

II) $V = \frac{1}{3}\pi \left(\frac{h}{6}\right)^2 h$

$V = \frac{1}{3}\pi \frac{h^3}{36}$

$V = \frac{\pi}{108} h^3$

$\frac{r}{10} = \frac{h}{60}$

$r = \frac{h}{6}$

III) $V = \frac{\pi}{108} h^3$

$\frac{dV}{dt} = \frac{\pi}{108} (3h^2 \frac{dh}{dt})$

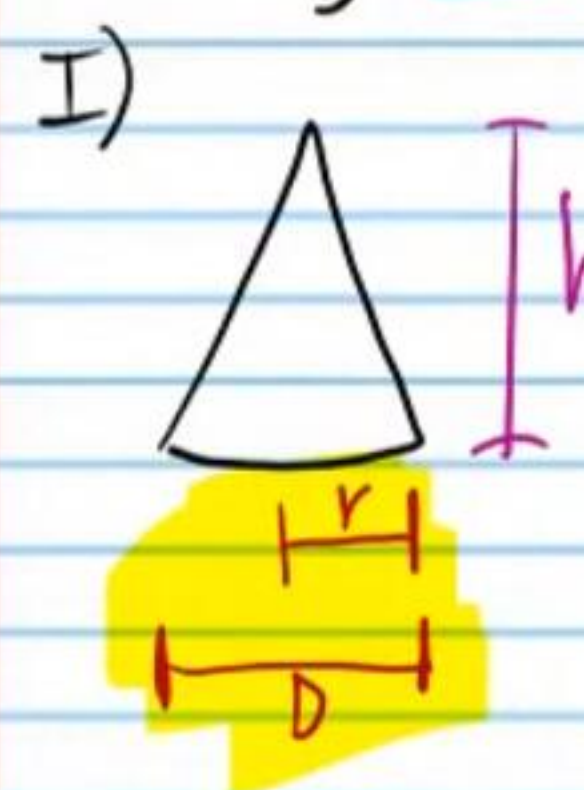
$\frac{dV}{dt} = \frac{\pi}{36} h^2 \frac{dh}{dt}$

IV) $\frac{dV}{dt} = \frac{\pi}{36} h^2 \frac{dh}{dt}$

$\frac{dV}{dt} = \frac{\pi}{36} (30^2) (-0.7)$

$\frac{dV}{dt} = -\frac{35\pi}{2} \text{ cm}^3/\text{s}$

⑧ Sand is falling off a conveyor belt and forming a pile in the shape of a cone. The height of the cone is always equal to the diameter of its base. The sand is added to the pile at $80 \text{ ft}^3/\text{h}$. Find the rate that the radius of the base of the pile is increasing when the height of the pile is 20 ft.



I) $V = \frac{1}{3}\pi r^2 h$

$V = \frac{1}{3}\pi r^2 (2r)$

$V = \frac{2\pi}{3} r^3$

II) $h = D$

$D = 2r$

$h = 2r$

III) $V = \frac{2\pi}{3} r^3$

$\frac{dV}{dt} = \frac{2\pi}{3} (3r^2 \frac{dr}{dt})$

$\frac{dV}{dt} = 2\pi r^2 \frac{dr}{dt}$

IV) $\frac{dV}{dt} = 2\pi r^2 \frac{dr}{dt}$

$80 = 2\pi (10^2) \frac{dr}{dt}$

$80 = 200\pi \frac{dr}{dt}$

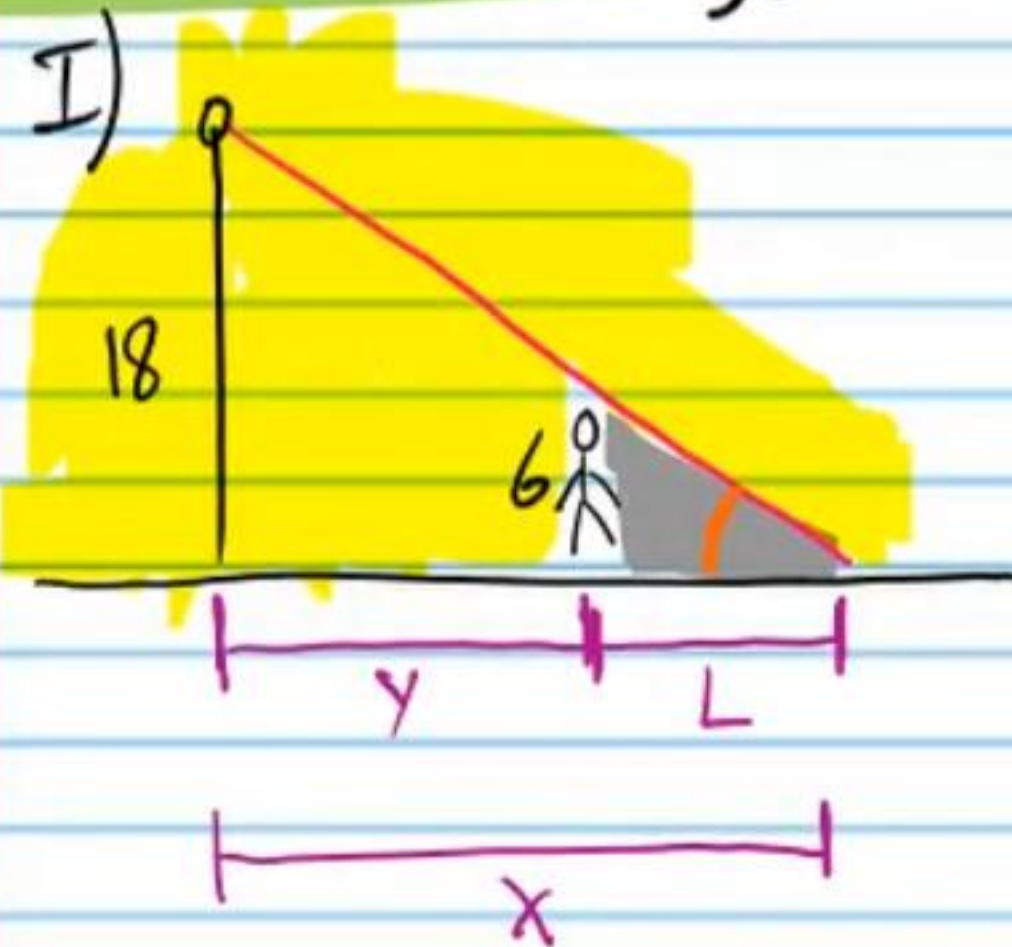
$\frac{2}{5\pi} \text{ ft/h} = \frac{dr}{dt}$

$h = 2r$

$20 = 2r$

$10 = r$

- ⑨ A man 6 ft tall walks away from a street light at a speed of 4 ft/s. The light is 18 ft above the ground. At what rate is the length of his shadow increasing?



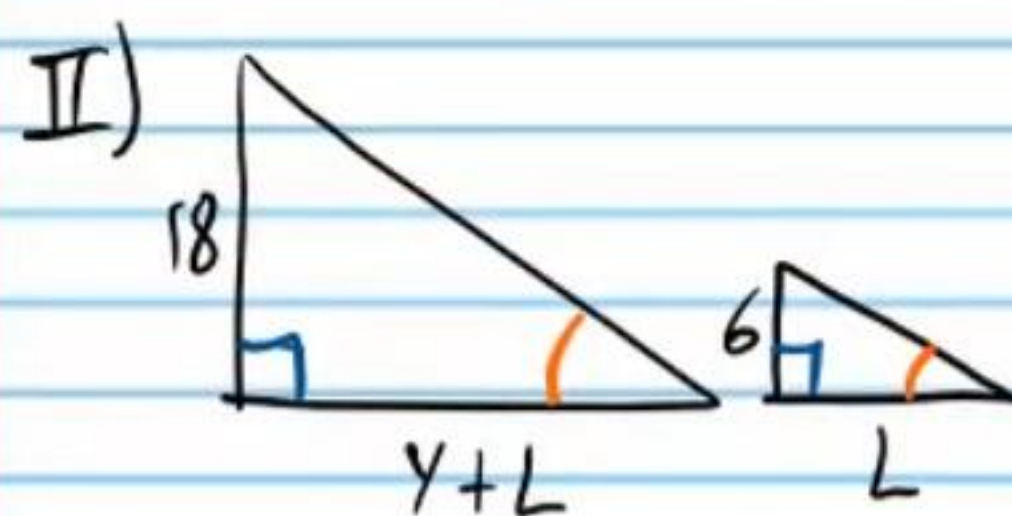
III) $2L = y$

$$2 \frac{dL}{dt} = \frac{dy}{dt}$$

IV) $2 \frac{dL}{dt} = \frac{dy}{dt}$

$$2 \frac{dL}{dt} = 4$$

$$\boxed{\frac{dL}{dt} = 2 \text{ ft/s}}$$



$$\frac{18}{6} = \frac{y+L}{L}$$

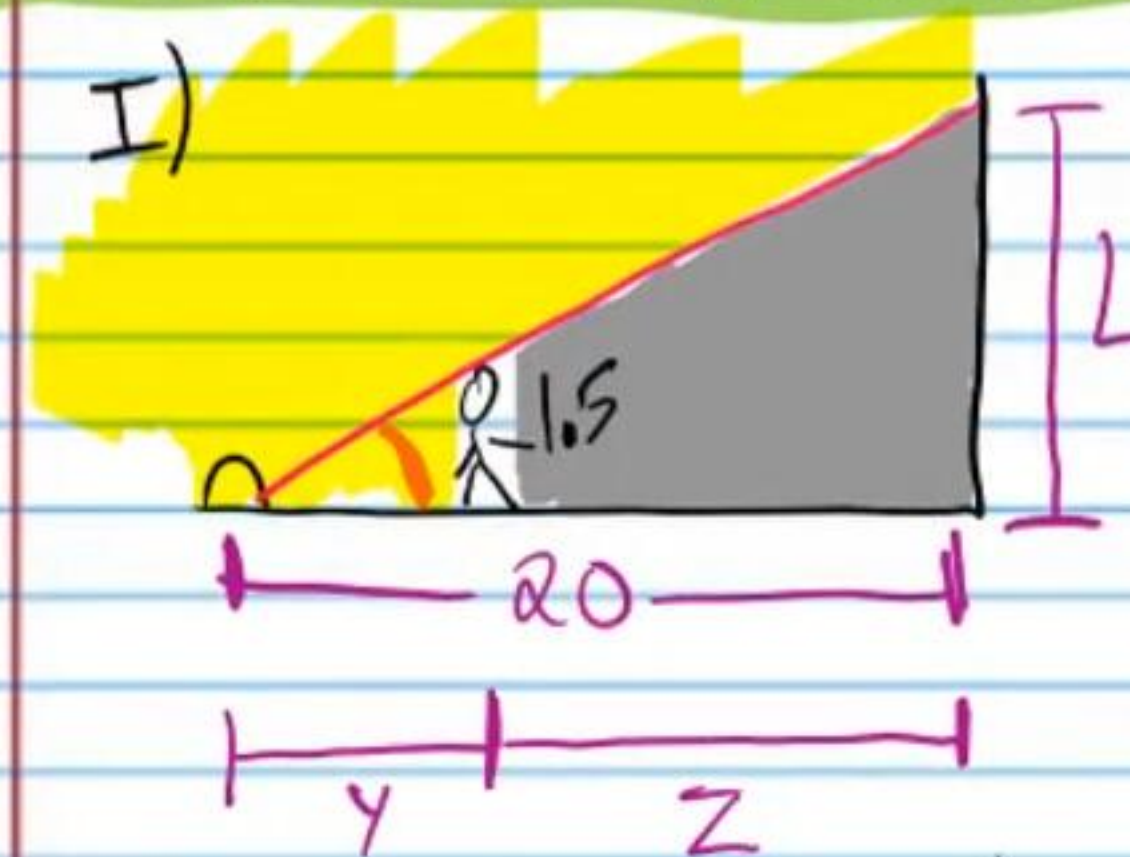
$$3 = \frac{y+L}{L}$$

$$(L) 3 = \frac{y+L}{L} (L)$$

$$3L = y+L$$

$$2L = y$$

- ⑩ A light on the ground shines on a wall 20 m away. If a woman 1.5 m tall walks from the light to the wall at a speed of 1 m/s, how fast is the length of her shadow on the wall decreasing when she is 2 m from the wall?

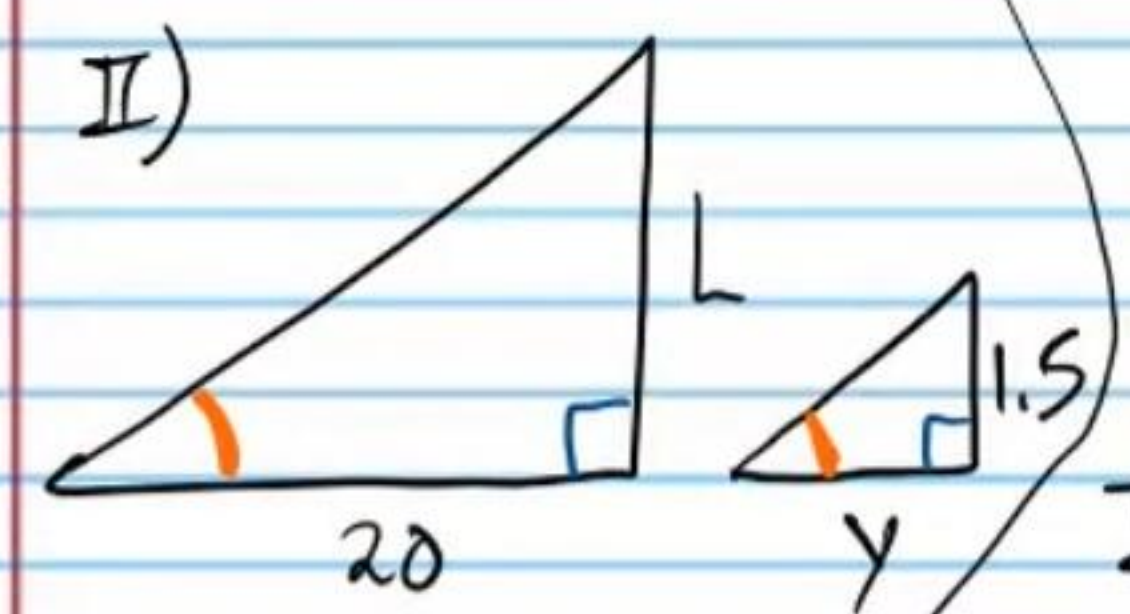


III) $L = \frac{30}{y}$

$$L = 30y^{-1}$$

$$\frac{dL}{dt} = 30(-y^{-2}) \frac{dy}{dt}$$

$$\frac{dL}{dt} = -\frac{30}{y^2} \frac{dy}{dt}$$



$$\frac{L}{1.5} = \frac{20}{y}$$

$$\frac{L}{\frac{3}{2}} = \frac{20}{y}$$

$$L = \frac{30}{y}$$

IV) $\frac{dL}{dt} = -\frac{30}{y^2} \frac{dy}{dt}$

$$\frac{dL}{dt} = -\frac{30}{18^2} (1)$$

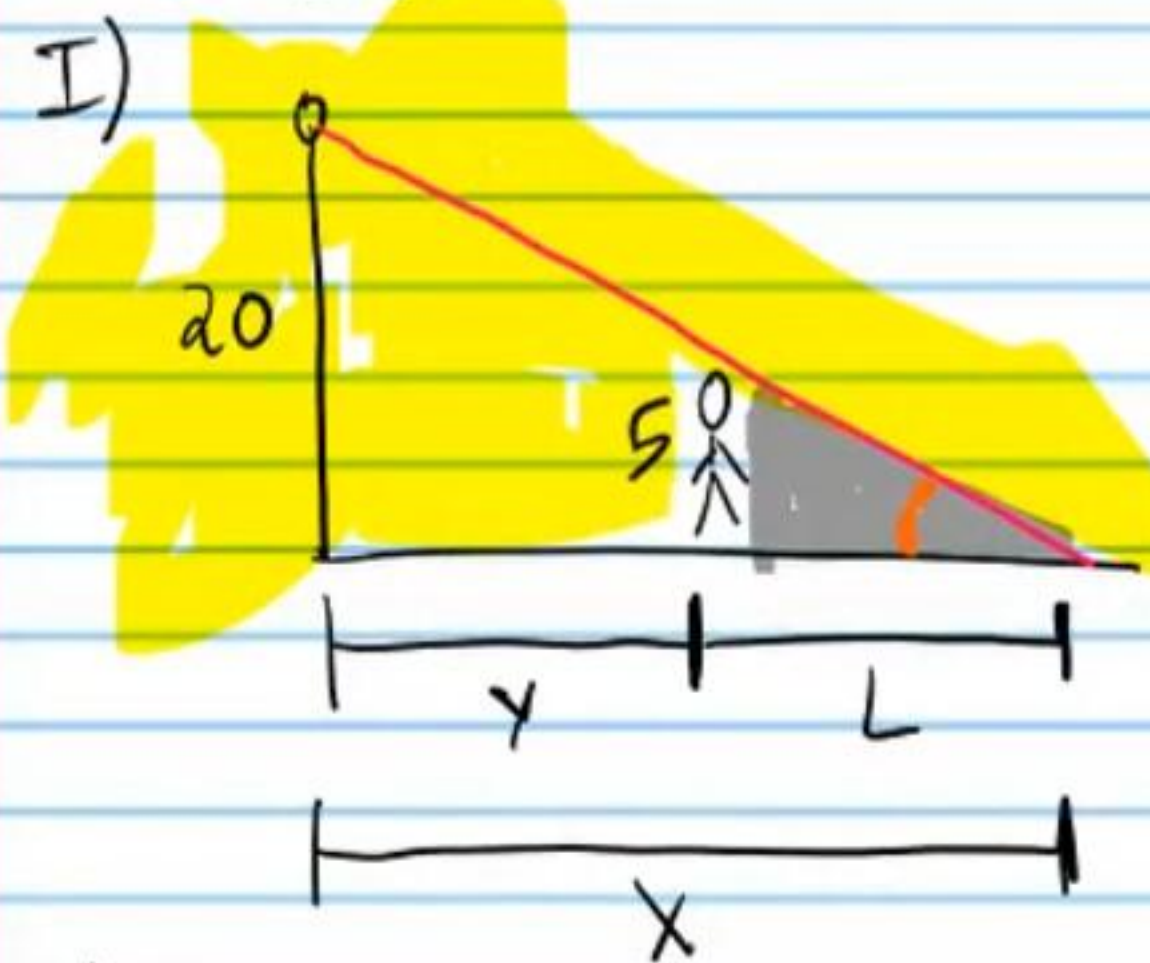
$$\boxed{\frac{dL}{dt} = -\frac{5}{54} \text{ m/s}}$$

$$20 = y+z$$

$$20 = y+2$$

$$18 = y$$

- (11) A person 5 ft tall walks toward a lamppost at a rate of 2 ft/s. The lamp is 20 ft above the ground. How fast is the length of the person's shadow decreasing?

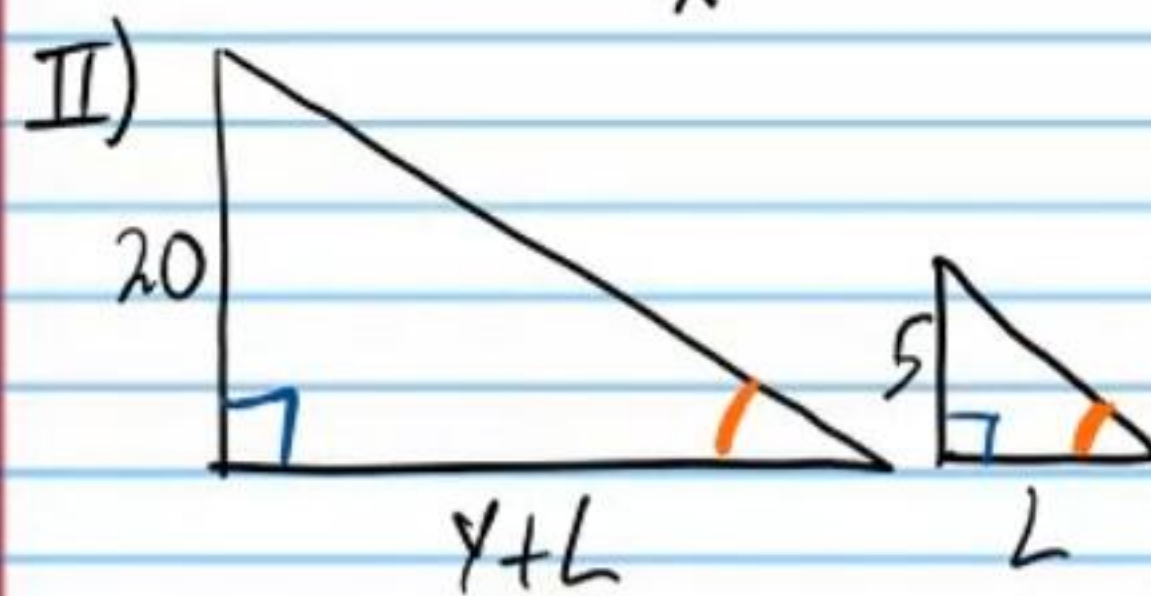


III) $3L = y$

$$3 \frac{dL}{dt} = \frac{dy}{dt}$$

$$3 \frac{dL}{dt} = -2$$

$$\boxed{\frac{dL}{dt} = -\frac{2}{3} \text{ ft/s}}$$



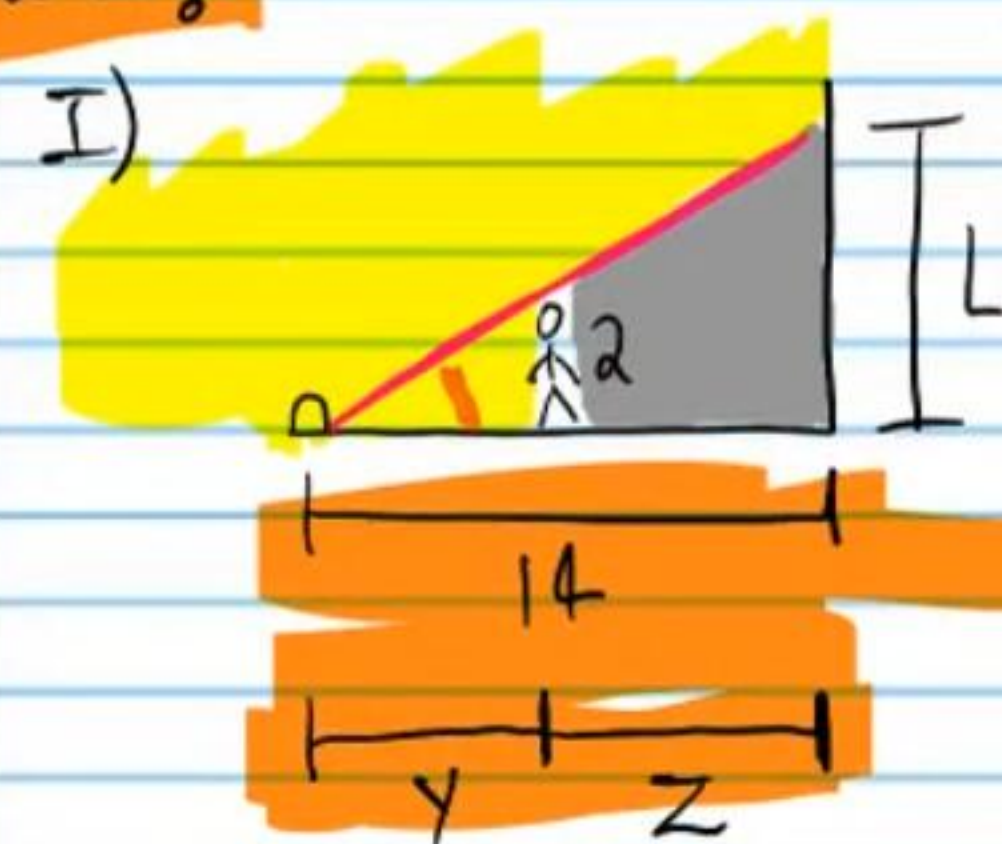
$$\frac{20}{5} = \frac{y+L}{L}$$

$$4 = \frac{y+L}{L}$$

$$4L = y+L$$

$$3L = y$$

- (12) A spotlight on the ground shines on a wall 14 meters away. A 2 m tall man walks from the wall to the light at 0.9 m/s. How fast is the length of his shadow on the wall increasing when he is 11 m from the wall?



III) $L = \frac{28}{y}$

$$L = 28y^{-1}$$

$$\frac{dL}{dt} = 28(-y^{-2} \frac{dy}{dt})$$

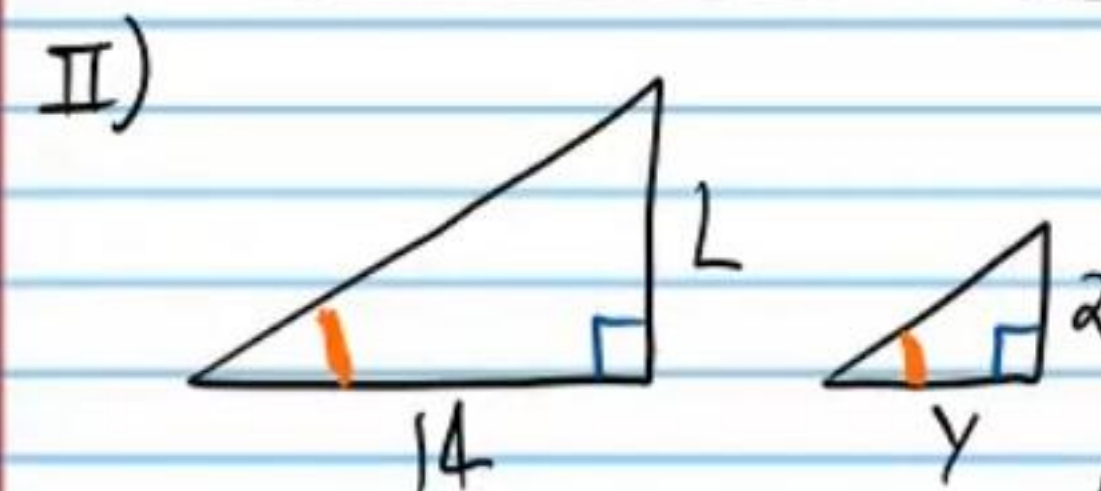
$$\frac{dL}{dt} = -\frac{28}{y^2} \frac{dy}{dt}$$

IV) $\frac{dL}{dt} = -\frac{28}{y^2} \frac{dy}{dt}$

$$\frac{dL}{dt} = -\frac{28}{3^2} (-0.9) \quad \begin{array}{l} 14 = y+2 \\ 14 = y+11 \\ 3 = y \end{array}$$

$$\frac{dL}{dt} = -\frac{28}{9} (-\frac{9}{10})$$

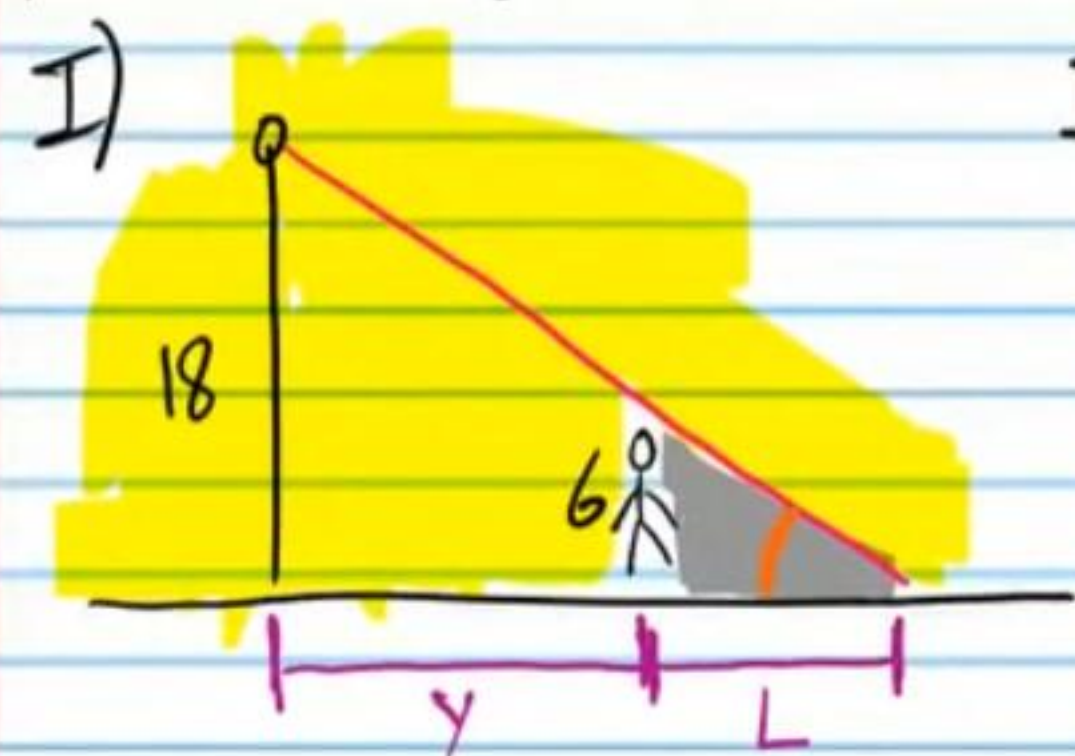
$$\boxed{\frac{dL}{dt} = 2.8 \text{ m/s}}$$



$$\frac{L}{2} = \frac{14}{y}$$

$$L = \frac{28}{y}$$

- (13) How fast is the tip of the shadow moving in problem #9?



II)

$$\begin{cases} 2L = y \\ 2(x-y) = y \\ 2x - 2y = y \\ 2x = 3y \end{cases} \quad \begin{cases} x = y + L \\ x - y = L \end{cases}$$

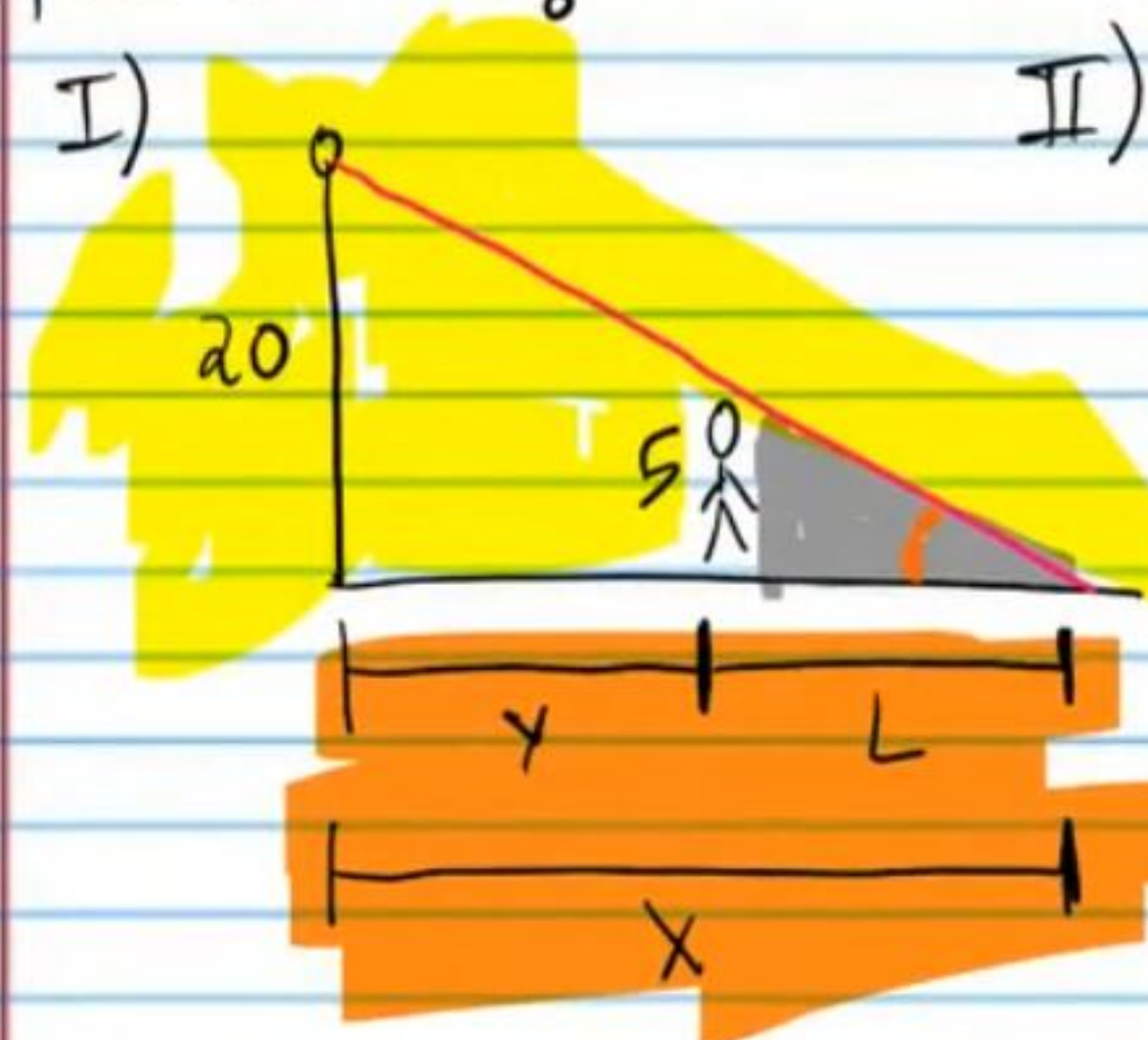
III)

$$2 \frac{dx}{dt} = 3 \frac{dy}{dt} \quad \text{IV)}$$

$$2 \frac{dx}{dt} = 3(4)$$

$$\boxed{\frac{dx}{dt} = 6 \text{ ft/s}}$$

- (14) At what rate is the tip of the shadow moving in problem #11?



II)

$$\begin{cases} 3L = y \\ 3(x-y) = y \\ 3x - 3y = y \\ 3x = 4y \end{cases} \quad \begin{cases} x = y + L \\ x - y = L \end{cases}$$

III)

$$3x = 4y$$

$$3 \frac{dx}{dt} = 4 \frac{dy}{dt}$$

IV)

$$3 \frac{dx}{dt} = 4 \frac{dy}{dt}$$

$$3 \frac{dx}{dt} = 4(-2)$$

$$\boxed{\frac{dx}{dt} = -\frac{8}{3} \text{ ft/s}}$$